

System Blueprint (*a.k.a.* "Design Doc")

TNPG: DeadDuckies

Project: Stardew Valley

Target ship date: 2026-06-01

Roster:

Name	Email	Primary Role	Secondary Role
Emaan Asif	emaana3@nycstudents.net	Project Manager	Inventory/Crafting
James Lei	jamesl291@nycstudents.net	Devo1	Entities/Player
Sarah Zou	sarahz40@nycstudents.net	Devo2	Farm/Town/Forest Map
Kiran Soemarjdo	kirans14@nycstudents.net	Devo3	Time/NPC's

Summary

We are creating a Stardew Valley emulator with our initial phase focused on the standard farm map. Games can be joined by players through the lobby. Players start with a farm with randomized trees and debris. Players will have inventories, be able to sell and buy goods, fish, mine, craft, and establish relationships along with a time and gold system.

Problem Being Solved

To offer a free, simplified, and web-version of Stardew Valley. Provides a trial of the game before users commit to buying the actual game.

Target Users

Users who enjoy Stardew Valley and farming games like it, or users who simply want to try out this genre of games.

Why This Project Matters

For us as developers, it builds off our work and lessons/skills learned from our previous P02 Civilization 6 emulator project while challenging us and helping us learn more about the gameplay logic and programming behind it. There are also many people we know who want to try Stardew Valley but are unable to or unsure about purchasing the actual game or do not have the storage for it.

Minimum Viable Product (MVP) Scope

Core Features (Required for Final Submission)

Features that **must** be completed:

1. Map
2. Tiles/Trees/Entities
3. Players
4. Movement
5. Inventory
6. Farming
7. Time
8. Stamina

Stretch Features (Only if MVP is Complete)

1. NPCs/Relationships
2. Mining/Fishing/Foraging
3. Crafting/Cooking/Construction/Tool Upgrades
4. Community Center
5. Skill Levels

Explicit Non-Goals

Features intentionally excluded:

- NPC schedules and pathways
- Festivals, animals, marriage, seasons and weather, and story events

Technology Stack

Layer	Selected Tool
Backend Framework	Flask
Frontend Framework	None
Database	SQLite
Authentication	Flask Sessions
ORM / DB Library	None

Why This Stack Was Chosen

This project is frontend heavy, with a focus on JavaScript and its canvas feature, we decided to choose the standard backend technology stack that we were all most familiar and comfortable in as well as to not rely on

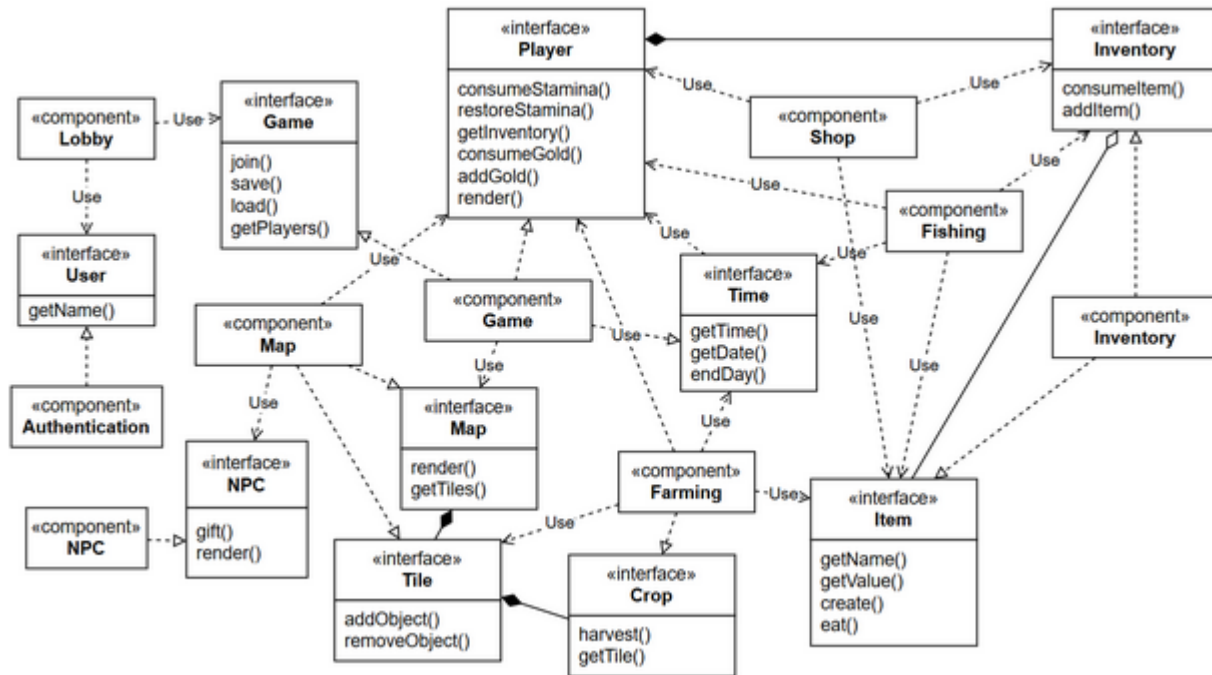
a frontend framework so we could have more flexibility.

Team Ownership Plan

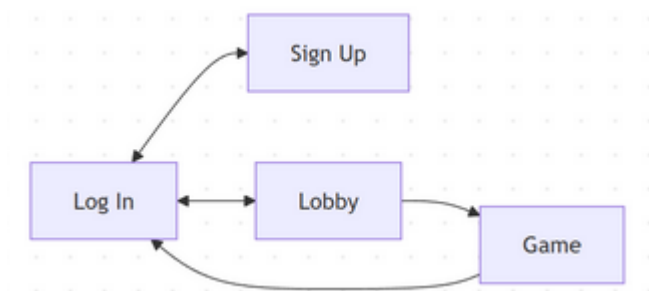
Each member must own meaningful deliverables.

Team Member	Primary Ownership	Secondary Ownership	Specific Deliverables
James	Player, Entities, Forest/Foraging, Mining	Houses, Static NPCs (Sebastian, Sam, Penny, Haley)	- Create movable player - Display multi-tile entities - Randomly spawn entities - Break entities to obtain items
Kiran	Time, Static NPCs (Willy, Pierre), Relationships, Fishing	Static NPCs (Sebastian, Sam, Penny, Haley), Community Center	- Create clock interface - Create NPC dialog - Track player relationships with NPCs - Option to gift NPC an item - Fishing minigame
Sarah	Maps, Pierre's, Travel, Stamina	Static NPCs (Gus, Clint)	- Mark passable and tillable tiles for Farm, Town, Quarry, and Forest - Player teleport between maps when walking on specific tiles - Create stamina bar - Render pierre's shop menu
Emaan	Inventory, Crafting, Cooking		- Display the hotbar, with hotkeys to switch slots - Open inventory menu, with the option to click and drag options - Crafting menu with recipe unlocks

Component Map



Site Map



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Landing Page
  ↓
Login / Register
  ↓
Lobby
  ├── Join Game
  ├── Load Save Game
  ↓
Game
  
```

Key User Stories

Farmer

As a farmer, I want to be tame the land in my farm, hoe tiles, and plant crops to make a nice big farm.

Monopolist

As a monopolist, I want to profit from the crops which I grow to obtain as much money as possible.

Hoarder

As a hoarder, I want to increase the amount of items I can store in my inventory to collect all the items in the game.

Database Design

Players

TEXT	name	UNIQUE
TEXT	password	

Games

INTEGER	id	PK AUTOINCREMENT
TEXT	player1	FK REF players(name)
TEXT	player2	FK REF players(name)
TEXT	player3	FK REF players(name)
TEXT	player4	FK REF players(name)
INTEGER	day	DEFAULT 0
INTEGER	money	DEFAULT 500

Tiles

INTEGER	game_id	FK REF games(id)
TEXT	map	
INTEGER	x_pos	
INTEGER	y_pos	
TEXT	furniture	DEFAULT ""
BOOLEAN	tiled	

Inventory

INTEGER	game	FK REF games(id)
TEXT	player	FK REF player(username)
TEXT	contents	
INTEGER	upgrades	DEFAULT 0

Plants

INTEGER	game_id	FK REF games(id)
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Plants

TEXT	map	
INTEGER	x_pos	
INTEGER	y_pos	
TEXT	type	
INTEGER	age	DEFAULT 0

Placeables

INTEGER	game_id	FK REF games(id)
TEXT	map	
INTEGER	x_pos	
INTEGER	y_pos	
TEXT	type	
INTEGER	item	
INTEGER	progress	

Relationships

INTEGER	game	FK REF games(id)
TEXT	npc_name	
TEXT	player	FK REF player(username)
INTEGER	points	
INTEGER	giftcount	DEFAULT 0
TEXT	status	

Testing Plan

- Lobby
 - Make sure that no more than 4 ppl can join a lobby, that the same player can't join twice, etc
 - Prevent duplicate results when multiple people perform the same action at the same time
- Map
 - Make sure all impassable tiles are indeed impassable
 - Make sure layers render correctly (eg, player is rendered behind objects he walks behind, and the objects fade properly)
- Authentication
 - Check that sqlite injection isn't possible
 - Ensure that there are no URLs that allow authentication to be bypassed
 - Make sure empty strings can't be used as username/password

- NPC
 - Ensure that dialogue triggers are correct
 - Make sure relationship system functions as intended
- Game
 - Ensure that saving works correctly
 - Check that no data is saved if the game is closed before the end of the day
- Farming
 - Make sure crop growth times are correct and that proper growth stages are correctly displayed
 - Ensure crops only grow when watered and can only be planted on tilled soil
 - Check that only tillable tiles can be hoed
- Shop
 - Ensure items are priced correctly
 - Players cannot go below 0 gold
- Fishing
 - Ensure fishing UI works as intended
- Inventory
 - Make sure items properly carry over between days
 - Make sure items can only be picked up if inventory has space

Timeline

Week 1 Goals:

- Farm Map (scrolling, software for json for impassable, untillable, teleporter tiles)
- Player Movement
- Entities (anything that goes on top of a tile, debris, worms, etc,)
- Time System
- Inventory

Week 2 Goals:

- Town Map
- Trees
- Pierre's
- Teleport/Travel

Week 3 Goals:

- Static NPCs (Willy, Pierre for fish and seed shops)
- Stamina System
- Forest Map/Foraging
- Mining

Week 4 Goals:

- Relationships
- Fishing

- Crafting
- Eating
- Cooking/Recipes
- Static NPCs (Gus, Clint for saloon and blacksmith)
- Static NPCs (Sebastian, Sam, Penny, Haley)
- Houses

CHOPPING BLOCK:

- Community Center
- Tool Upgrades
- Skill Levels
- Sound Effects/Music

Internal Deadlines:

1. Farm map and player movement (5/6/2026)
2. Entities, main game UI and time/inventory system (5/11/2026)
3. Trees (5/13/2026)
4. Town map (5/13/2026)
5. Teleport (5/13/2026)
6. Pierre's (5/15/2026)
7. Static NPCs (Willy, Pierre) (5/16/2026)
8. Stamina (5/18/2026)
9. Forest Map/Foraging (5/25/2026)
10. Static NPCs (Gus, Clint) (6/01/2026)
11. Mining (6/01/2026)
12. Eating (6/14/2026)
13. Relationships (6/14/2026)
14. Crafting (6/14/2026)
15. Fishing (6/14/2026)
16. Houses (6/14/2026)
17. Static NPCs (Sebastian, Sam, Penny, Haley) (6/14/2026)
18. Cooking/Eating (6/14/2026)

Completion Criteria (*a.k.a.* "Definition of 'Done'")

Project is considered complete when all of the following are true:

1. Players are able to move and interact with the map to obtain resources
2. Players are able to farm and fish
3. Players are able to buy and sell items

Open Questions

- Implement community center/skill levels?

- Multiplayer?

Appendix

Assets:

- <https://lybell-art.github.io/xnb-js/> (used to extract files from Stardew Valley)